Q.1 What is need for Hypothesis Testing ?

Ans:- Hypothesis Testing is a fundamental statistical technique used to make decisions about a population based on sample data. In real-world scenarios, it is often impractical or impossible to study an entire population due to constraints such as time, cost, and accessibility. Therefore, hypothesis testing provides a scientific and systematic approach to draw conclusions using sample observations.

The primary need for hypothesis testing arises in the following areas:

1. Decision Making Under Uncertainty

In many practical situations, decisions must be made with incomplete information. Hypothesis testing allows us to make rational decisions by evaluating evidence from sample data.

2. Validation of Assumptions

Researchers and analysts often propose assumptions or claims regarding population parameters. Hypothesis testing helps in verifying whether these assumptions are statistically valid or not.

3. Scientific Research

It plays a crucial role in research by testing theories, models, and experimental results. It ensures that conclusions are not based on random chance but on statistical evidence.

4. Quality Control and Industrial Applications

Industries use hypothesis testing to maintain quality standards. For example, testing whether the average weight of a product meets the specified requirement.

5. Comparative Studies

It is used to compare two or more groups, such as testing the effectiveness of a new drug versus an existing one.

Thus, hypothesis testing is essential for making logical, data-driven decisions in various fields such as science, engineering, business, and healthcare.

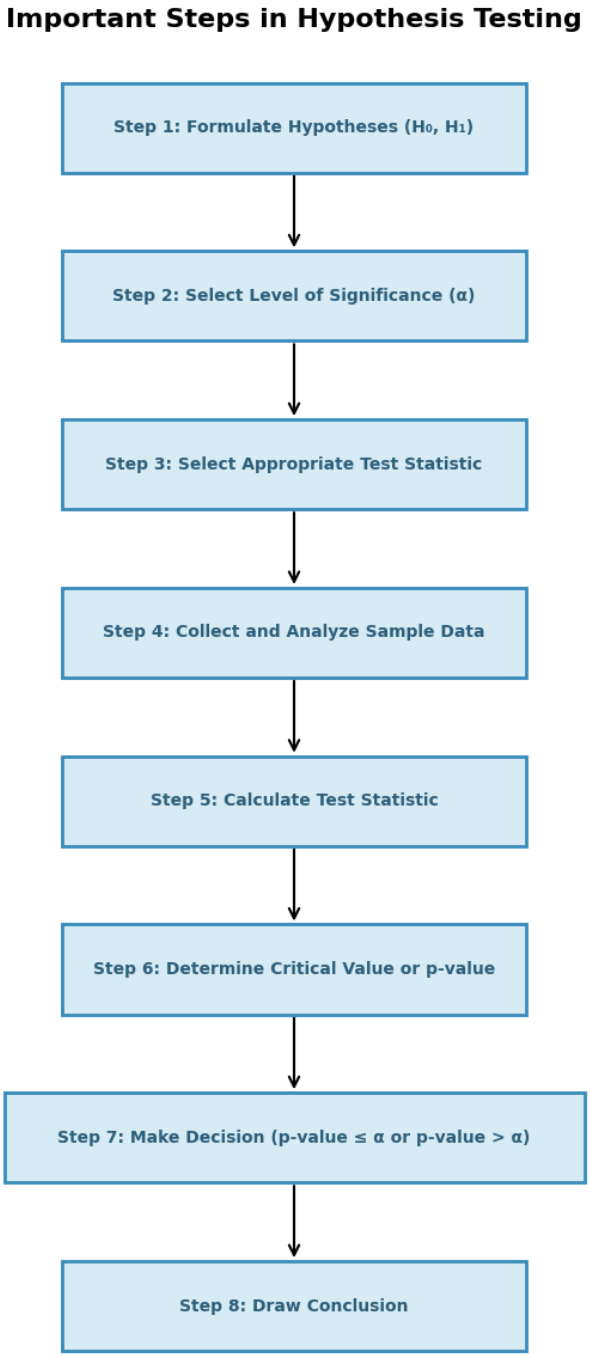
Q.2 With a neat diagram show important steps in hypothesis Testing.

Ans:-

```
import matplotlib.pyplot as plt
import matplotlib.patches as patches

# Define the steps
steps = [
```

```
    "Step 1: Formulate Hypotheses (H₀, H₁)",
    "Step 2: Select Level of Significance (α)",
    "Step 3: Select Appropriate Test Statistic",
    "Step 4: Collect and Analyze Sample Data",
    "Step 5: Calculate Test Statistic",
    "Step 6: Determine Critical Value or p-value",
    "Step 7: Make Decision (p-value ≤ α or p-value > α)",
    "Step 8: Draw Conclusion"
]

# Setup the figure and axes
fig, ax = plt.subplots(figsize=(10, 12))
ax.set_xlim(0, 10)
ax.set_ylim(0, 12)
ax.axis('off') # Hide the axes

# Define box properties
box_width = 4
box_height = 0.8

# Define starting position
start_x = 3
start_y = 11

# Draw boxes and add text for each step
boxes = []
for i, step_text in enumerate(steps):
    y_pos = start_y - i * (box_height + 0.7)
    if i == 6: # Special handling for decision step (making it slightly wide
        current_box_width = box_width + 1
        current_box_x = start_x - 0.5 # Adjust x to keep it centered
    else:
        current_box_width = box_width
        current_box_x = start_x

    rect = patches.Rectangle((current_box_x, y_pos), current_box_width, box_
                             facecolor='#D9EDF7', edgecolor='#3A8DBC', linew
    ax.add_patch(rect)
    ax.text(current_box_x + current_box_width / 2, y_pos + box_height / 2, s
            ha='center', va='center', fontsize=10, weight='bold', color='#2F
    boxes.append((current_box_x, y_pos, current_box_width, box_height))

# Draw arrows to connect the steps
for i in range(len(boxes) - 1):
    x1, y1, w1, h1 = boxes[i]
    x2, y2, w2, h2 = boxes[i+1]

    # Arrow from bottom center of current box to top center of next box
    ax.annotate('', xy=(x2 + w2/2, y2 + h2), xytext=(x1 + w1/2, y1),
                arrowprops=dict(facecolor='black', edgecolor='black', arrows
                ha='center', va='center', zorder=1)

# Specific arrows for the decision step (Step 7)
x_decision, y_decision, w_decision, h_decision = boxes[6]
x_conclusion, y_conclusion, w_conclusion, h_conclusion = boxes[7]
```

```
# Arrow for Reject H₀ (left path from decision)
# For simplicity, I'm just showing a single path into the conclusion for thi
# as the previous text cell describes the two conditions leading to a single

plt.title('Important Steps in Hypothesis Testing', fontsize=16, weight='bold
plt.tight_layout()
plt.show()
```



Q.3 What is a student t distribution?

Ans:- The Student's t-distribution is a probability distribution used in statistical analysis when the sample size is small and the population standard deviation is unknown.

It was introduced by William Sealy Gosset under the pseudonym "Student."

Characteristics of t-Distribution:

- It is bell-shaped and symmetrical, similar to the normal distribution.
- It has heavier tails, which means more variability.
- It depends on degrees of freedom (df = n - 1).
- As sample size increases, the t-distribution approaches the normal distribution.

Applications:

- Used in t-tests to estimate population mean.
- Suitable when sample size is less than 30.
- Used when population standard deviation is unknown.

Thus, the t-distribution is an important tool for making reliable inferences from small sample data.

Q.4 What is level of significance ? What role t plays in hypothesis Testing

Ans:- Level of Significance (α)

The level of significance is defined as the probability of rejecting a true null hypothesis. It represents the risk of committing a **Type I error**.

Common values include:

- α = 0.05 (5% level of significance)
- α = 0.01 (1% level of significance)

A lower value of α indicates stricter criteria for rejecting the null hypothesis.

Role of t in Hypothesis Testing

The t-statistic is used to determine how far the sample mean deviates from the population mean in terms of standard error.

Where:

- $\bar{x}$ x ¯ = sample mean
- μ μ = population mean
- s s = sample standard deviation
- n n = sample size

Importance of t-value:

- Helps in determining statistical significance.
- Larger t-value indicates stronger evidence against H_0.
- Used to compute p-value and compare with critical values.

Q.5 Design a hypothesis testing for identifying average age of students in TE IT class is 19 or not.

Ans:- Problem Statement:

To test whether the average age of TE IT students is 19 years or not.

Step 1: Formulate Hypotheses

- H_0: $\mu = 19$
- H_1: $\mu \neq 19$

(This is a two-tailed test)

Step 2: Choose Level of Significance

- $\alpha = 0.05$ (95% confidence level)

Step 3: Select Test

Since the sample size is small and population standard deviation is unknown, a one-sample t-test is used.

Step 4: Sample Data (Example)

18, 19, 20, 19, 21, 18, 19, 20

Step 5: Calculate Sample Statistics

- Mean ($\bar{x}$) ≈ 19.25
- Standard deviation (s) ≈ 1.03
- Sample size (n) = 8

Step 6: Compute t-value

Using the t-formula, we calculate the test statistic.

Step 7: Decision Rule

- Degrees of freedom = 7
- Critical t-value (two-tailed, $\alpha = 0.05$) ≈ ±2.365

Step 8: Conclusion

- If calculated t lies within the range → Accept H_0
- Otherwise → Reject H_0

In this case, the calculated value is within the acceptable range, so the null hypothesis i

Q.6 Check if you can accept Ho at alpha 95% ? Match your conclusions with python functions.

Ans:- Conclusion:

At a 95% confidence level, there is no significant difference between the sample